

Bluestock Mutual Fund Analytics Capstone Project

Executive Summary

This project analyses the Indian Mutual Fund industry using data engineering, data analytics, risk analysis, and business intelligence techniques. The objective was to build an end-to-end analytics solution that transforms raw mutual fund datasets into actionable insights for investors and financial institutions.

The project includes ETL pipelines, data cleaning, exploratory data analysis, performance analytics, advanced risk metrics, and an interactive Power BI dashboard.

1. Introduction

The Indian Mutual Fund industry has experienced significant growth in assets under management (AUM), investor participation, and SIP contributions. Understanding fund performance, investor behaviour, and market trends is essential for informed investment decisions.

The objective of this project is to analyse mutual fund schemes, benchmark performance, investor transactions, and market trends using modern data analytics techniques.

2. Project Objectives

1. Build a complete ETL pipeline.
2. Store processed data in SQLite.
3. Perform exploratory data analysis.
4. Calculate advanced performance metrics.
5. Develop an interactive dashboard.
6. Conduct advanced investor analytics.
7. Generate actionable insights and recommendations.

3. Data Sources

The project uses multiple datasets:

- Mutual Fund NAV History
- Scheme Performance Data
- Industry AUM Statistics
- SIP Inflows
- Category Inflows
- Investor Transactions
- Portfolio Holdings
- Nifty Benchmark Indices

These datasets were cleaned, transformed, and integrated into a unified analytics environment.

4. ETL Pipeline Design

The ETL process consists of:

Extract

Raw CSV files were collected from publicly available mutual fund datasets.

Transform

- Missing value handling
- Data type corrections
- Date standardisation
- Duplicate removal
- Feature engineering

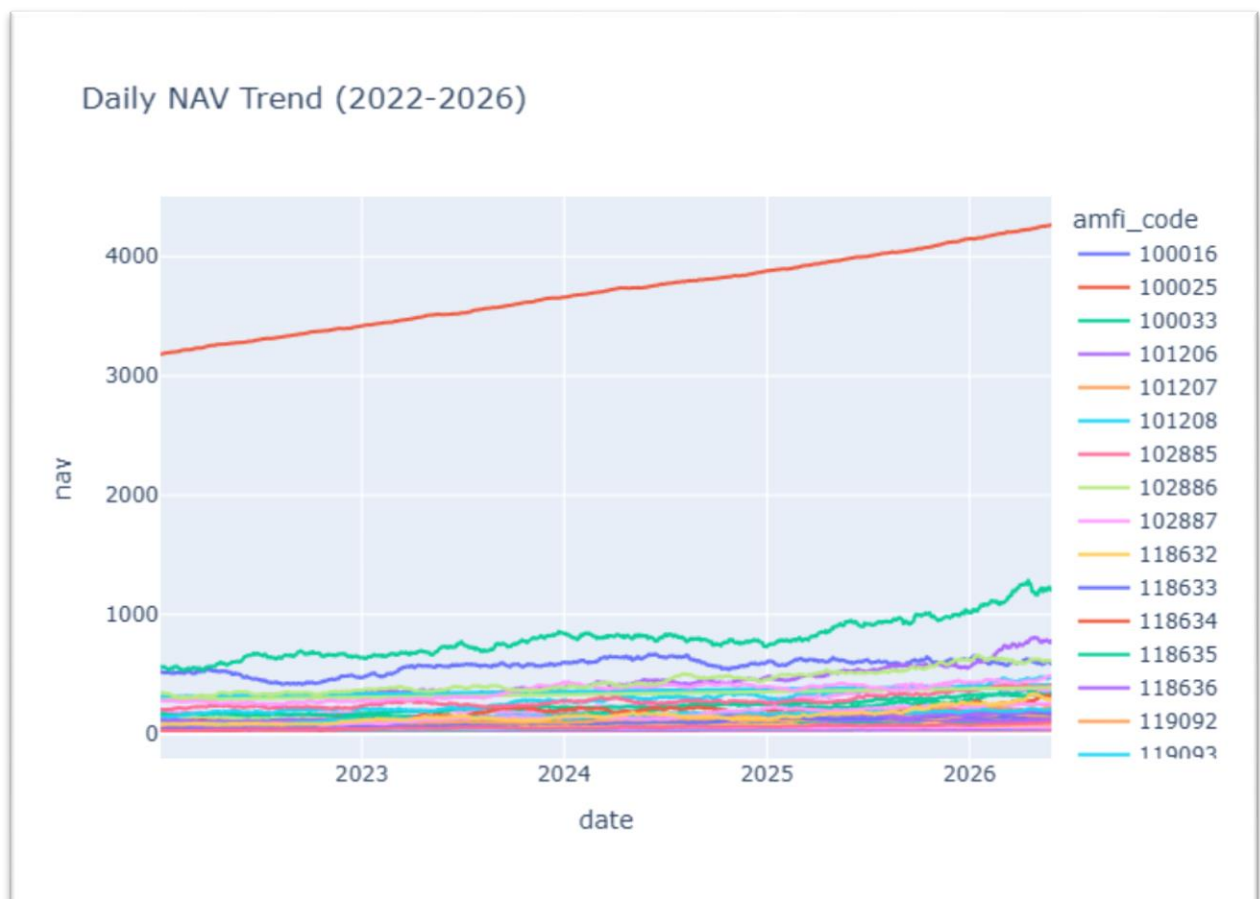
Load

Processed datasets were loaded into SQLite and exported as cleaned CSV files for analytics and dashboarding.

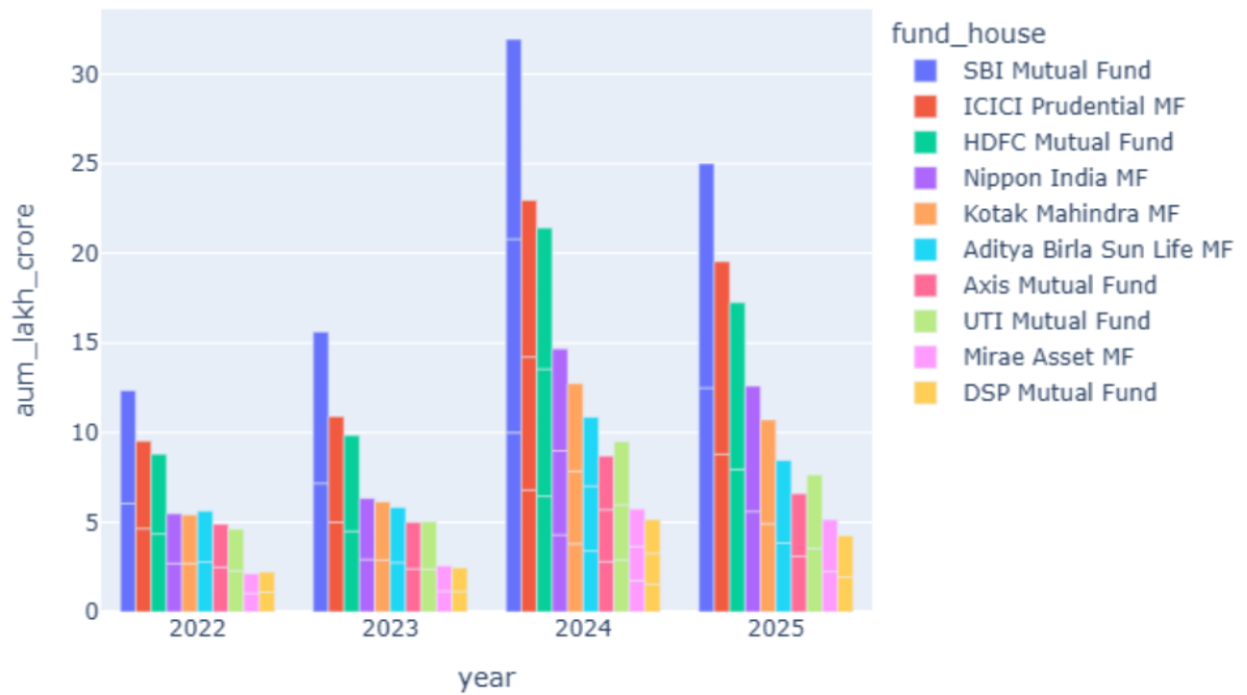
5. Exploratory Data Analysis

Key analyses performed:

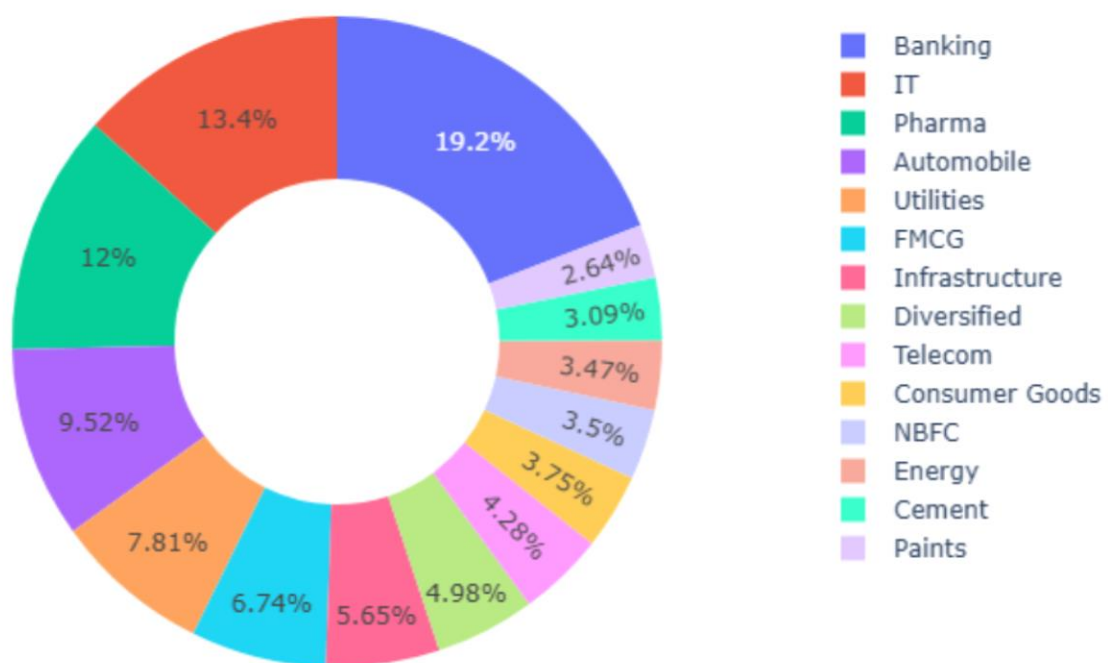
- Industry AUM growth trend
- SIP inflow trend
- Fund house comparison
- Folio growth analysis
- Sector allocation analysis
- Demographic analysis
- State-wise investor distribution



AUM Growth by Fund House (2022-2025)



Sector Allocation Across Equity Funds



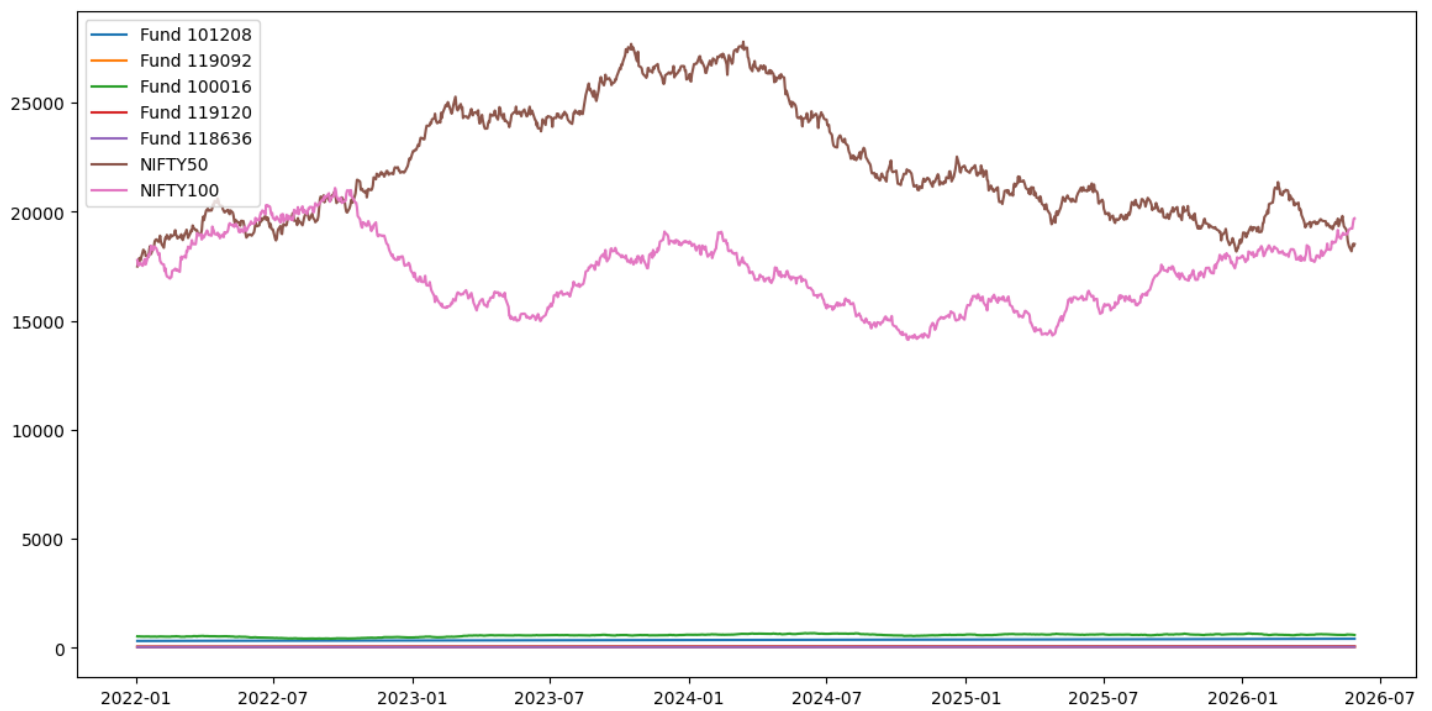
Monthly SIP Inflow Trend (2022-2025)



6. Performance Analytics

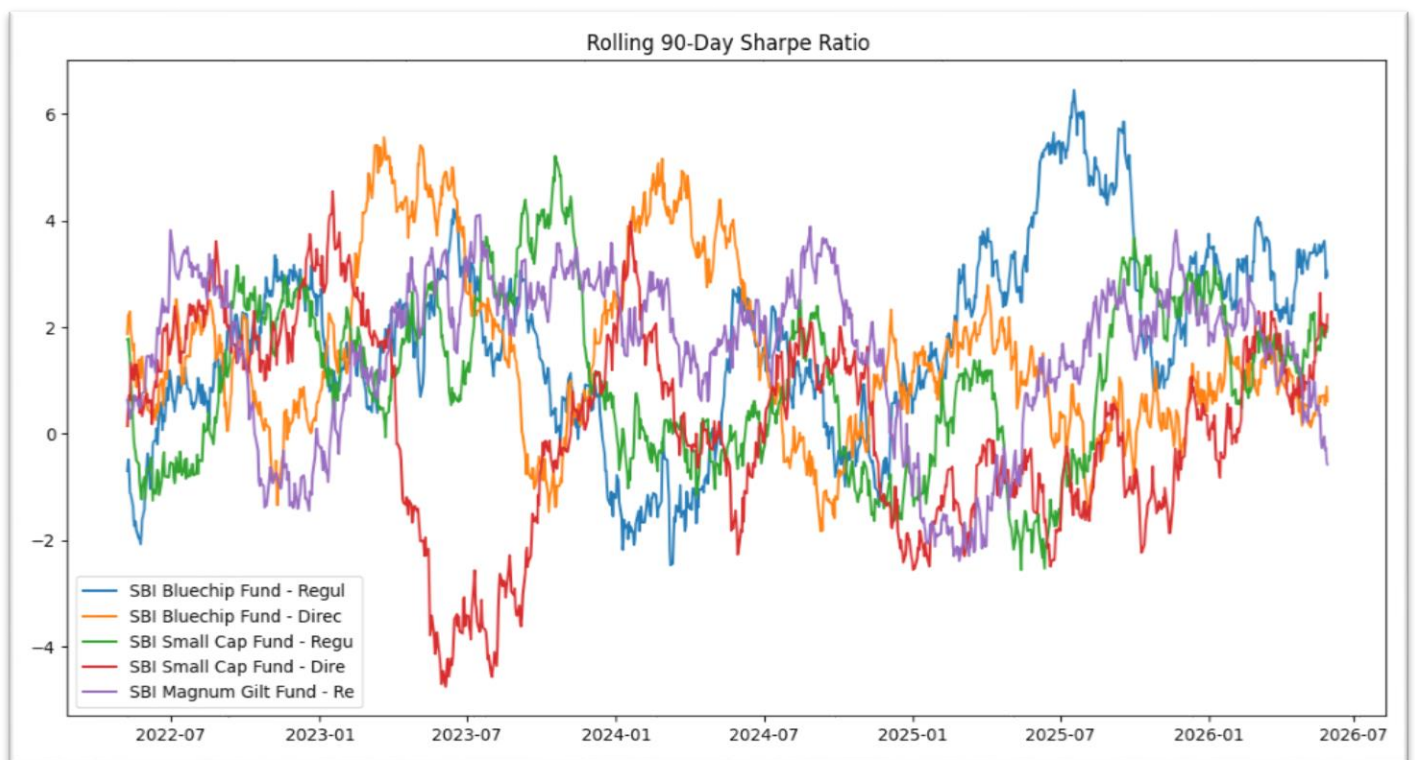
Metrics calculated:

- Daily Returns
- CAGR (1Y, 3Y, 5Y)
- Sharpe Ratio
- Sortino Ratio
- Alpha
- Beta
- Maximum Drawdown
- Fund Scorecard
- Tracking Error



Advanced risk and investor analyses include:

- Historical VaR
- Conditional VaR
- Rolling Sharpe Ratio
- Investor Cohort Analysis
- SIP Continuity Analysis
- Sector HHI Concentration
- Fund Recommendation Engine



8. Dashboard Development

Four interactive dashboard pages were developed:

Industry Overview

KPIs, AUM trend, AMC comparison

Fund Performance

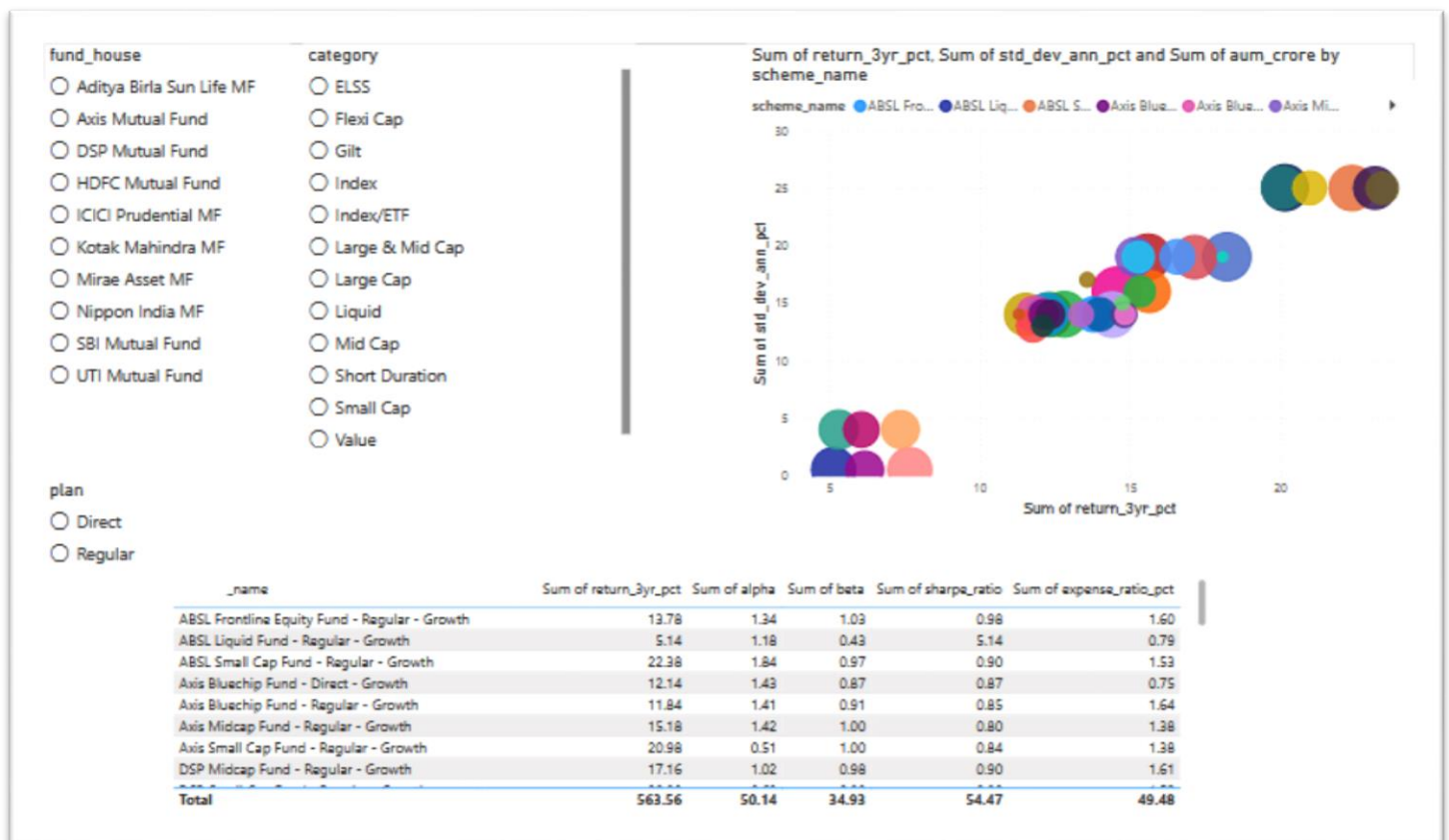
Risk-return scatter plot, scorecards, NAV analysis

Investor Analytics

State, age, and transaction insights

SIP & Market Trends

Category inflows, SIP trends, benchmark trends



Total AUM (₹ Lakh Cr)

SIP Inflows (₹ Cr)

Total Folios (Cr)

Total Schemes

391.76

Sum of aum_lakh_crore

940K

Sum of sip_inflow_crore

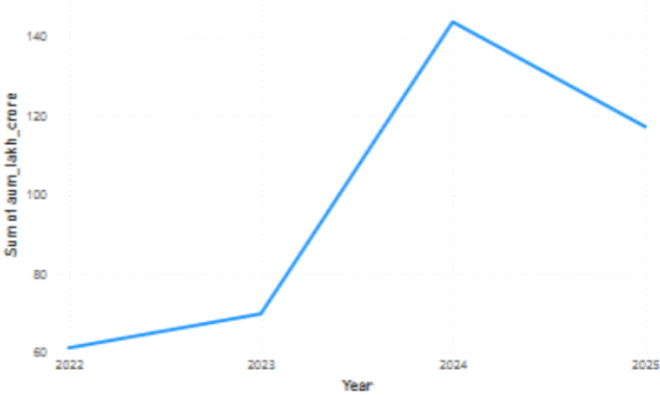
416.42

Sum of total_folios_crore

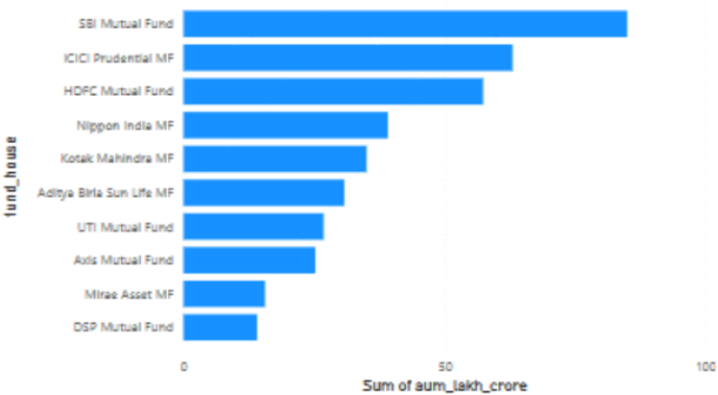
40

Count of scheme_name

Industry AUM Trend (2022-2025)



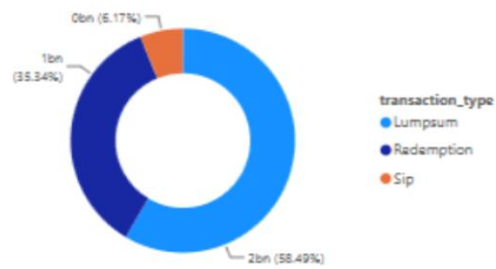
AUM by Fund House



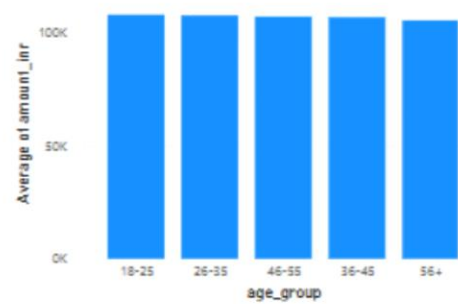
state

- ☐ Delhi
- ☐ Gujarat
- ☐ Haryana
- ☐ Karnataka
- ☐ Madhya Pradesh
- ☐ Maharashtra
- ☐ Punjab
- ☐ Rajasthan
- ☐ Tamil Nadu
- ☐ Telangana
- ☐ Uttar Pradesh
- ☐ West Bengal

Transaction Type Distribution



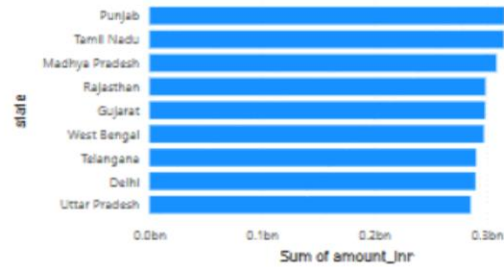
Average Investment by Age Group



age_group

- ☐ 18-25
- ☐ 26-35
- ☐ 36-45
- ☐ 46-55
- ☐ 56+

Transaction Amount by State



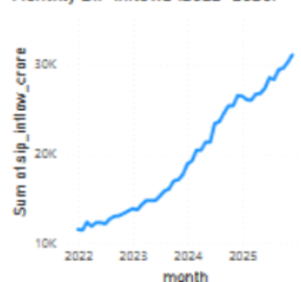
Monthly Transaction Volume



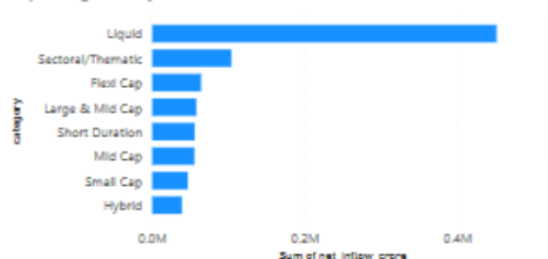
city_tier

- ☐ B30
- ☐ T30

Monthly SIP Inflows (2022–2025)



Top Categories by Net Inflow



month

- ☐ 01 April 2024
- ☐ 01 May 2024
- ☐ 01 June 2024
- ☐ 01 July 2024
- ☐ 01 August 2024
- ☐ 01 September 2024
- ☐ 01 October 2024
- ☐ 01 November 2024
- ☐ 01 December 2024
- ☐ 01 January 2025
- ☐ 01 February 2025
- ☐ 01 March 2025

category	01 April 2024	01 May 2024
ELSS	466	
Flexi Cap	4947	
Gilt	784	
Hybrid	2955	
Large & Mid Cap	4214	
Large Cap	2413	
Liquid	37537	4
Mid Cap	3897	
Sectoral/Thematic	8052	
Short Duration	4400	
Small Cap	3533	
Total	74526	8

9. Key Findings

1. SIP inflows consistently increased across the study period.
2. Equity-focused funds generated stronger long-term returns.
3. Risk-adjusted performance varied significantly across schemes.
4. Investor participation increased steadily across cohorts.
5. Several funds displayed concentrated sector exposure.

10. Limitations

- Historical data limitations.
- Market conditions may change future performance.
- Simplified recommendation logic.
- Limited benchmark coverage.

11. Recommendations

- Investors should focus on risk-adjusted returns.
- SIP continuity should be monitored.
- Diversification can reduce concentration risk.
- Dashboard monitoring should be updated periodically.

12. Conclusion

This project successfully demonstrates the complete analytics lifecycle from raw data ingestion to business intelligence reporting. The developed dashboard and analytics framework provide valuable insights into mutual fund performance, investor behaviour, and industry trends.